Competition for Feedstock Among Four Plastic-Recycling Technologies

A generalized Nash equilibrium model of gasification, gasification with CCS, STRAP and incineration

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1 Introduction

1.1 The problem

The multilayer-film decision-support paper treats waste management as a *planning* problem: one decision maker has a fixed amount of post-industrial multilayer film and chooses how much to send to each treatment route so as to maximize total profit. That model answers "what is the best allocation?" It does not say what the waste is *worth*, and it does not describe a market, where each treatment plant is run by a separate firm that buys waste to maximize its own profit.

This report studies that market. Four technologies compete for the same limited supply of waste:

- **Gasification**, producing hydrogen;
- **Gasification + CCS**, the same process with carbon capture, which earns the U.S. Section 45Q carbon-capture tax credit;
- **STRAP**, the solvent-based recycling process for multilayer film (its residue is landfilled);
- **Incineration**, with energy recovery.

Every ton one firm buys is a ton the others cannot have. The central question is: *what price does competition put on a ton of waste, who ends up buying it, what does this depend on, and does each plant earn back its capital?*

1.2 Approach

We model the firms as players in a *generalized Nash equilibrium problem* (GNEP): each firm maximizes its own profit, but all firms share one constraint, the total waste available. We compute the *variational equilibrium*, in which all firms face the same price on the shared constraint. That price, called λ (USD per ton), is the main output. It is not assumed; it is the shadow price of the shared waste constraint. Each firm's profit includes its annualized capital charge, and we report whether each plant covers it.

1.3 Summary of results

For a base case of 20,000 t/yr of waste (Section 3):

- The equilibrium price is $\lambda = 932.96$ USD/ton, and **STRAP** sets it.
- Gasification + CCS and Gasification each take their full 8,000 t/yr, and STRAP takes the remaining 4,000 t/yr (half its capacity). Incineration takes nothing.
- With the 45Q credit (270 USD/t), Gasification + CCS earns almost exactly the same margin as plain Gasification (1,410.77 against 1,410.00): the credit offsets the extra 269 USD/t of capture cost. The two are effectively tied.
- **STRAP does not earn back its capital at this pool size.** It runs half empty and loses 2.58 million USD/yr after its annualized capital; it needs to process 6,768 t/yr (85% of capacity) to break even. Both gasification plants cover their capital.
- The price is a staircase in the amount of waste: five values, switching at 8,000, 16,000, 24,000 and 36,800 t/yr. The base case sits 4,000 t/yr inside each edge of its step.
- At the base case the price depends one-for-one on STRAP's own revenue and operating cost, on the feedstock price and on the plant capacities, and not on the other firms' numbers (within wide limits).

1.4 Conventions

Every input is tagged **[PAOLA]** (number supplied by Paola) or **[ASSUMED]** (our default, to be confirmed). Each result is followed by a green **Interpretation** box that says in plain words what it means. Orange boxes mark something that could change a conclusion. The questions for Paola are collected in Section 4.3.

2 Method

2.1 Setting and notation

Let $\mathcal{I} = \{\text{Gas}, \text{CCS}, \text{STRAP}, \text{Inc}\}$ be the set of firms. All economic quantities are per ton of feedstock and constant in the amount processed (linear cost), as supplied by Paola.

2.2 Each firm's problem

Firm i chooses q_i to maximize its profit after capital

$$\max_{q_i \geq 0} \quad \pi_i = m_i\, q_i - \text{ACC}_i \qquad \text{s.t.} \qquad q_i \leq \text{Cap}_i, \qquad \sum_{j \in \mathcal{I}} q_j \leq Q_{\text{tot}}. \tag{1}$$

The last constraint is the same for all firms: it is the shared supply of waste. Because it involves every firm's decision, this is a *generalized* Nash game and not an ordinary one. The capital charge ACC_i is in the objective, but it is a constant: it does not depend on q_i, so it does not change the optimal throughput. It shifts profit, and it is what we use to decide whether a plant pays back (Section 2.7).

Table 1: Notation.

Symbol	Meaning
q_i	Tons per year of waste bought and processed by firm i (decision variable).
Cap_i	Plant capacity of firm i, t/yr: the most it can process.
r_i	Revenue of firm i per ton of feed, USD/t. For the CCS firm it includes the 45Q credit.
a_i	Operating cost of firm i per ton of feed, USD/t.
ACC_i	Annualized capital charge of firm i, USD/yr.
p_{raw}	Price the waste generator charges per ton (0 in the base case).
m_i	Net margin of firm i: $m_i = r_i - a_i - p_{\mathrm{raw}}$, USD/t. The most firm i can pay for one more ton and still break even on operations.
Q_{tot}	Total waste available, t/yr.
λ	Multiplier of the shared waste constraint, USD/t: the equilibrium price of waste.
μ_i	Multiplier of firm i's capacity constraint, USD/(t/yr): the gain in firm i's operating profit per extra ton/yr of its own capacity.

2.3 Equilibrium concept

We solve for the *variational equilibrium*: the equilibrium in which every firm attaches the same multiplier λ to the shared constraint. This is the natural concept for a market with one clearing price, and it is the one for which λ can be read as "the price of waste". Attach $\mu_i \geq 0$ to $q_i \leq \mathrm{Cap}_i$ and $\lambda \geq 0$ to the shared constraint. Stacking the optimality (KKT) conditions of all four firms gives

$$0 \leq q_i \ \perp \ \lambda + \mu_i - m_i \geq 0, \tag{2}$$

$$0 \leq \mu_i \ \perp \ \mathrm{Cap}_i - q_i \geq 0, \tag{3}$$

$$0 \leq \lambda \ \perp \ Q_{\mathrm{tot}} - \textstyle\sum_j q_j \geq 0, \tag{4}$$

for all $i \in \mathcal{I}$. The notation $x \perp y$ means $xy = 0$. The term $\lambda + \mu_i - m_i$ in (2) does not depend on q_i. That is the formal statement that cost is linear, and it drives everything below.

2.4 Solution

Best response. By (2)–(3), each firm's best response to a price λ is a step:

$$q_i(\lambda) = \begin{cases} \mathrm{Cap}_i & \text{if } m_i > \lambda, \\ 0 & \text{if } m_i < \lambda, \\ \text{any value in } [0, \mathrm{Cap}_i] & \text{if } m_i = \lambda. \end{cases} \tag{5}$$

A firm that earns more per ton than the price wants as much as its plant can process, and a firm that earns less wants none.

Aggregate demand and the price. Total demand at price λ is the step function $D(\lambda) = \sum_{i:\, m_i > \lambda} \mathrm{Cap}_i$. By (4) the equilibrium price is the lowest price at which demand fits the supply:

$$\lambda^* = \min\{\lambda \geq 0 : \ D(\lambda) \leq Q_{\mathrm{tot}}\}. \tag{6}$$

Firms strictly above λ^* run at capacity, firms strictly below take nothing, and a firm exactly at λ^* takes whatever is left. At the exact thresholds where D steps, any price between two adjacent steps is consistent with equilibrium; (6) picks the lowest and we report that.

Closed form (merit order). Sort the firms by margin, $m_{(1)} > m_{(2)} > \dots$, and let $K_k = \mathrm{Cap}_{(1)} + \cdots + \mathrm{Cap}_{(k)}$ be the running total of their capacities. Then

$$\lambda^*(Q_{\mathrm{tot}}) = m_{(k)} \quad \text{for } K_{k-1} \le Q_{\mathrm{tot}} < K_k \qquad (K_0 = 0), \tag{7}$$

and $\lambda^* = 0$ once Q_{tot} exceeds the capacity of every firm with a positive margin. In words: the firms form a queue by margin, each takes its full capacity, and the first firm that the supply cannot fill completely is the price-setter. The price is a staircase in Q_{tot} with steps at the K_k. No numerical solver is needed.

2.5 Equivalence with the planner's problem

The conditions (2)–(4) are exactly the KKT conditions of the linear program

$$\max_{q \ge 0} \sum_{i \in \mathcal{I}} m_i q_i \quad \text{s.t.} \quad q_i \le \mathrm{Cap}_i, \quad \sum_i q_i \le Q_{\mathrm{tot}}, \tag{8}$$

with λ and μ_i as its dual variables (adding the constants $-\mathrm{ACC}_i$ to the objective does not change the solution). This gives three facts that we use in the results:

1. Competition does not change *who* gets the waste: the equilibrium allocation equals the planner's. What the game adds is the price λ.
2. $\mu_i = m_i - \lambda$ for a firm at capacity. For two firms i, j, the gain from moving one ton from j to i is $m_i - m_j$.
3. By LP strong duality, total operating profit splits exactly as

$$\sum_i m_i q_i^* \;=\; \lambda^* Q_{\mathrm{tot}} + \sum_i \mu_i^* \,\mathrm{Cap}_i. \tag{9}$$

   The first term is the value of the pile at the clearing price. The second is *capacity rent*: the extra margin earned by firms that are full and cannot grow.

2.6 Equilibrium states

Table 2: The three states a firm can be in at equilibrium.

State	Condition	Meaning
At capacity	$m_i > \lambda$, $q_i = \mathrm{Cap}_i$, $\mu_i > 0$	Earns more per ton than the price and would take more waste if it could.
Price-setter	$m_i = \lambda$, $0 < q_i < \mathrm{Cap}_i$	Only partly filled. Indifferent about one more ton, and its margin *is* the price.
Priced out	$m_i < \lambda$, $q_i = 0$	Margin below the price; takes nothing.

2.7 Viability: does a plant earn back its capital?

The equilibrium treats capacities as given, so a plant that exists runs whenever it earns positive cash. Whether it would have been *built* is a separate question. We answer it with two quantities for each firm that has a capital charge:

$$\pi_i^* = m_i q_i^* - \mathrm{ACC}_i, \qquad q_i^{\mathrm{BE}} = \frac{\mathrm{ACC}_i}{m_i}. \tag{10}$$

π_i^* is the profit after capital at equilibrium and q_i^{BE} is the break-even throughput: the tons per year the plant must process to cover its capital. A plant is viable if $\pi_i^* \geq 0$, that is, if $q_i^* \geq q_i^{\mathrm{BE}}$. This is a consistency check on the equilibrium, not part of it: a firm with $\pi_i^* < 0$ would not build, which changes the game (Section 3.3).

2.8 Inputs

All margins come from Paola's table (revenue and operating cost per ton of feed, and annualized capital for an 8,000 t/yr plant). Points about how the table is used:

- STRAP's operating cost is labelled "with landfill", so the landfilling of its residue is already included and no separate landfill fee is charged. Nothing from Gasification or Incineration goes to landfill. Waste that no firm buys is landfilled, which costs the firms nothing.
- The CCS firm earns the U.S. Section 45Q credit for capturing and storing carbon (Global CCS Institute, 2025[1]), which Paola puts at 270 USD per ton of feed. We add it to the CCS revenue: 1,635 + 270 = 1,905 USD/t.
- Capital is annualized capital in million USD per year, as supplied. It is not supplied for Incineration.

Table 3: Inputs. Revenue and operating cost in USD per ton of feed; capital in million USD per year.

Firm	Revenue r_i	Cost a_i	Capacity	Capital	Source
Gasification	1,635.00	224.9987	8,000	5.9998	**[PAOLA]**; cap. **[ASSUMED]**
Gasification + CCS	1,905.00	494.2344	8,000	7.0775	**[PAOLA]**; cap. **[ASSUMED]**
STRAP	1,468.386	535.4272	8,000	6.3145	**[PAOLA]**; cap. **[ASSUMED]**
Incineration	259.5664	138.0971	12,800	not supplied	**[PAOLA]**; cap. **[ASSUMED]**

Capital = ACC_i in million USD/yr. CCS revenue = 1,635 plus 270 for the 45Q credit. STRAP's cost includes landfill. Shared: $Q_{\mathrm{tot}} = 20{,}000$ t/yr **[ASSUMED]**; $p_{\mathrm{raw}} = 0$ USD/t **[ASSUMED]**.

What the capacities mean. A capacity is the most a plant can process per year. We set each plant at the scale Paola's cost was quoted for: 8,000 t/yr for Gasification, Gasification + CCS

[1] https://www.globalccsinstitute.com/u-s-preserves-and-increases-45q-credit-in-one-big-beautiful-bill-act/

and STRAP (the scale of the capital figures too), and 12,800 t/yr for Incineration (its cost is for a 100,000 t/yr plant, of which 12.8% is plastic). These are our assumptions and Paola should confirm them (Q2). The waste pool of 20,000 t/yr is also an assumption (Q1): it is larger than any single plant, so that the firms genuinely compete.

What is not in the model. There are no emission factors, so emissions are not reported (Q9). Prices of products are fixed, and costs are constant per ton (Section 4.1).

3 Results

3.1 Net margins

Table 4: Net margin $m_i = r_i - a_i - p_{\text{raw}}$, USD/ton, in queue order.

Firm	Revenue	− Operating cost	= Net margin m_i
Gasification + CCS	1,905.00	494.23	**1,410.77**
Gasification	1,635.00	225.00	**1,410.00**
STRAP	1,468.39	535.43	**932.96**
Incineration	259.57	138.10	**121.47**

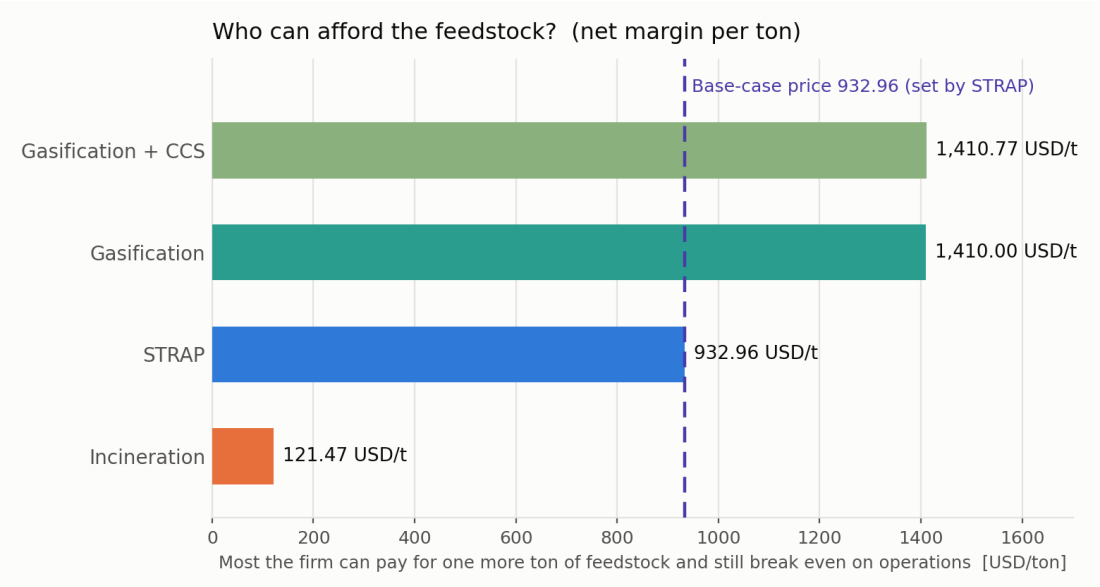


Figure 1: Each firm's walk-away price. The dashed line is the equilibrium price at 20,000 t/yr: both gasification routes are above it, STRAP sits exactly on it, and Incineration is far below.

Interpretation

Gasification and Gasification + CCS (1,410). The two highest margins, and almost identical. Without the credit, carbon capture would add 269 USD/t of operating cost (+120%) for no revenue and leave CCS at 1,140.77, 270 USD/t behind. The 270 USD/t credit cancels that extra cost to within 0.76 USD/t, so CCS comes out marginally *ahead* (1,410.77 against 1,410.00). Both beat STRAP by about 477 USD/t.
STRAP (932.96). STRAP sells resin at a high price, but has the highest operating cost of the four (535). It ranks third, which is what makes it the price-setter at 20,000 t/yr.
Incineration (121.47). Energy is its only product. Its margin is one-eighth of STRAP's, so it is the last resort: it gets waste only after every other plant is full.

The two gasification routes are nearly tied

Which one is ahead rests on 0.76 USD/t. If the credit were any smaller than 269.24 USD/t, plain Gasification would be first in the queue. This does not move the base-case price, because both plants are full and STRAP sets it. It only decides which of the two sets the price when the pool is below 16,000 t/yr, and there the two candidate prices differ by 0.76 USD/t. It is, however, why the credit's exact value and basis are worth confirming (Q6).

3.2 Base case: 20,000 t/yr

Filling the queue. Applying (7), the firms line up by margin:

1. Gasification + CCS takes its full 8,000 t/yr. 12,000 t/yr remain.
2. Gasification takes its full 8,000 t/yr. 4,000 t/yr remain.
3. STRAP has room for 8,000 t/yr but only 4,000 remain. It is half filled and exactly on the margin, so it is the price-setter.
4. Incineration gets nothing: the supply is used up.

The price is therefore STRAP's margin:

$$\boxed{\lambda^* = m_{\text{STRAP}} = 1{,}468.39 - 535.43 - 0 = 932.96\ USD/\text{ton}}$$

Table 5: Base-case equilibrium. Operating profit is margin × tons.

Firm	Tons/yr	Share	State	Margin	μ_i	Operating profit (USD/yr)
Gasification + CCS	8,000	40%	At capacity	1,410.77	477.81	11,286,125
Gasification	8,000	40%	At capacity	1,410.00	477.04	11,280,011
STRAP	4,000	20%	Price-setter	932.96	0	3,731,835
Incineration	0	0%	Priced out	121.47	0	0
Total	20,000	100%				26,297,971

Interpretation

Four firms, three states (Table 2). The two gasification plants are at capacity: they would take more waste if they could. STRAP is the price-setter: half-filled, and the price is whatever makes it indifferent. Incineration is priced out. The states carry the economics: "at capacity" means wants more, "price-setter" means indifferent, "priced out" means wants none.

What $\lambda = 932.96$ means. It is the price of scarcity, and it is also the most a waste generator could charge. If the generator charged p_{raw} per ton, each firm's margin would fall by p_{raw} and λ would fall by the same amount, so the total a firm pays, $p_{\text{raw}} + \lambda$, stays at 932.96. At $p_{\text{raw}} = 100$ the price is 832.96 and all 20,000 t/yr are still sold. At 932.96 and above STRAP drops out and 4,000 t/yr go unsold; Gasification drops out at 1,410.00 and Gasification + CCS at 1,410.77, after which nobody buys.

Interpretation

932.96 is the generator's *reservation price*. In the base case the generator gives the waste away, so nobody pays λ, but the model says the generator gives up as much as 932.96 USD/t, about USD18.7 million per year on 20,000 t. Setting $p_{\text{raw}} = 0$ is therefore not neutral: it hands the whole price of scarcity to the recyclers. A positive p_{raw} would share it between generator and recyclers without changing the total cost to a recycler, as long as all the waste is still sold. Section 3.3 shows that the plants could not actually afford to pay all of it.

Where the profit comes from. By (9),

$$\underbrace{26{,}297{,}971}_{\text{operating profit}} = \underbrace{\lambda^* Q_{\text{tot}}}_{932.96\times 20{,}000 = 18{,}659{,}177} + \underbrace{\sum_i \mu_i \text{Cap}_i}_{3{,}822{,}454 + 3{,}816{,}340 = 7{,}638{,}794} .$$

The shadow prices are $\mu_{\text{CCS}} = m_{\text{CCS}} - m_{\text{STRAP}} = 477.81$ and $\mu_{\text{Gas}} = m_{\text{Gas}} - m_{\text{STRAP}} = 477.04$ USD/(t/yr).

Interpretation

About 71% of the operating profit is the value of the pile at the clearing price, and 29% is capacity rent earned by the two gasification plants. STRAP, as the price-setter, earns no rent: its whole profit ($932.96 \times 4{,}000 = 3.73$ million) is just the pile price on the tons it takes. The rent goes to the plants that are full, not to the one that sets the price.
$\mu_{\text{Gas}} = 477.04$ also means that moving one ton/yr of capacity from STRAP to Gasification would add 477.04 USD/yr to the system's operating profit; the same move to Gasification + CCS adds 477.81, and to Incineration nothing.

Not included in this result

Emissions. Without per-ton emission factors we cannot report system emissions (Q9). Capital is treated in the next subsection.

3.3 Profit after capital

Using (10) with the annualized capital in Table 3:

Table 6: Profit after capital at 20,000 t/yr. Incineration's capital was not supplied, so it is not shown.

Firm	Operating profit	Annualized capital	Profit after capital	Break-even tons/yr	Break-even % of capacity
Gasification + CCS	11,286,125	7,077,509	4,208,615	5,017	63%
Gasification	11,280,011	5,999,788	5,280,222	4,255	53%
STRAP	3,731,835	6,314,456	−2,582,620	6,768	85%

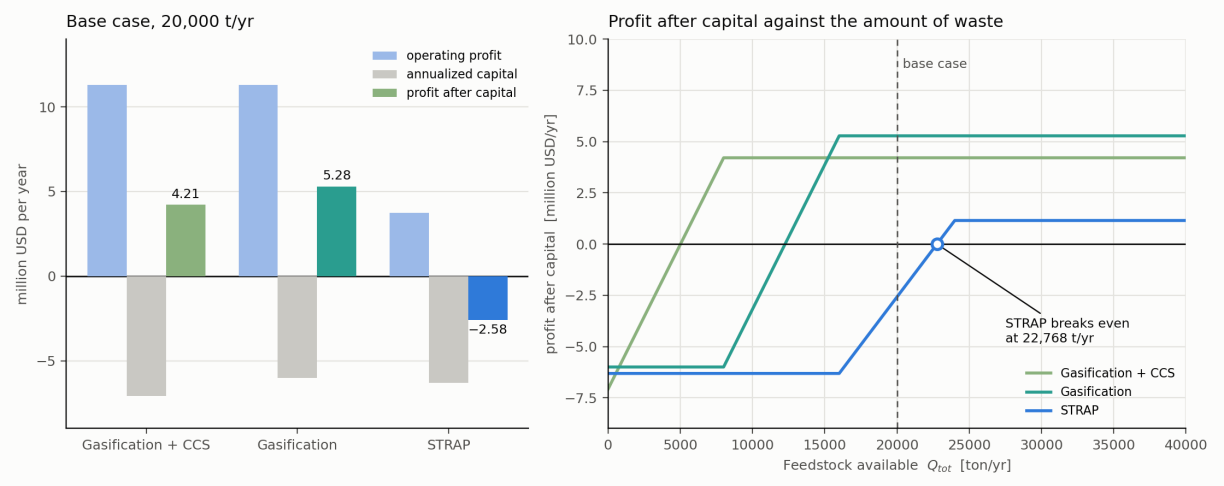


Figure 2: Left: operating profit, annualized capital and profit after capital at the base case. Right: profit after capital as the amount of waste varies. Each curve is flat once its plant is full. STRAP only breaks even once the pool reaches 22,768 t/yr.

Interpretation

The two gasification plants pay back comfortably. Each needs only about half to two-thirds of its capacity to cover its capital, and each is full.

STRAP does not. It is the price-setter, so it gets only the 4,000 t/yr left over, half its plant. Its operating profit of 3.73 million does not cover its 6.31 million capital charge, a loss of 2.58 million USD/yr. STRAP needs 6,768 t/yr, which in the queue means a pool of at least 16,000 + 6,768 = 22,768 t/yr. Between 22,768 and 24,000 t/yr STRAP covers its capital *and* the price is still 932.96.

What happens if STRAP is not built. The pool of 20,000 t/yr is then split 8,000 / 8,000 / 4,000 between the two gasification plants and Incineration, and the price falls to Incineration's margin, 121.47 USD/t (an 87% drop). The gasification plants' profits do not change, because the waste is free to them.

The generator cannot capture the whole price. Because the gasification plants' capacity rent (about 3.8 million each) is less than their capital charge (6.0 and 7.1 million), they only pay back because the waste is free. Each ton of feedstock price takes 8,000 USD/yr off a full plant. Gasification stays viable up to $p_{\text{raw}} = 660$ USD/t and Gasification + CCS up to 526 USD/t, well below the 932.96 reservation price.

How to read this

The equilibrium of Section 3.2 takes the plants as already built. The loss for STRAP is a warning that, at 20,000 t/yr and these capacities, STRAP would not have invested; it is not a change to the equilibrium itself. Incineration's capital is not supplied (Q8), so the "STRAP not built" case treats the incinerator as already built. Also, the break-even figures use the margin with the waste free; a positive feedstock price raises each plant's break-even throughput.

3.4 Price against the amount of waste

We re-solved the equilibrium for pools from 0 to 40,000 t/yr (209 points, including each threshold and points just either side).

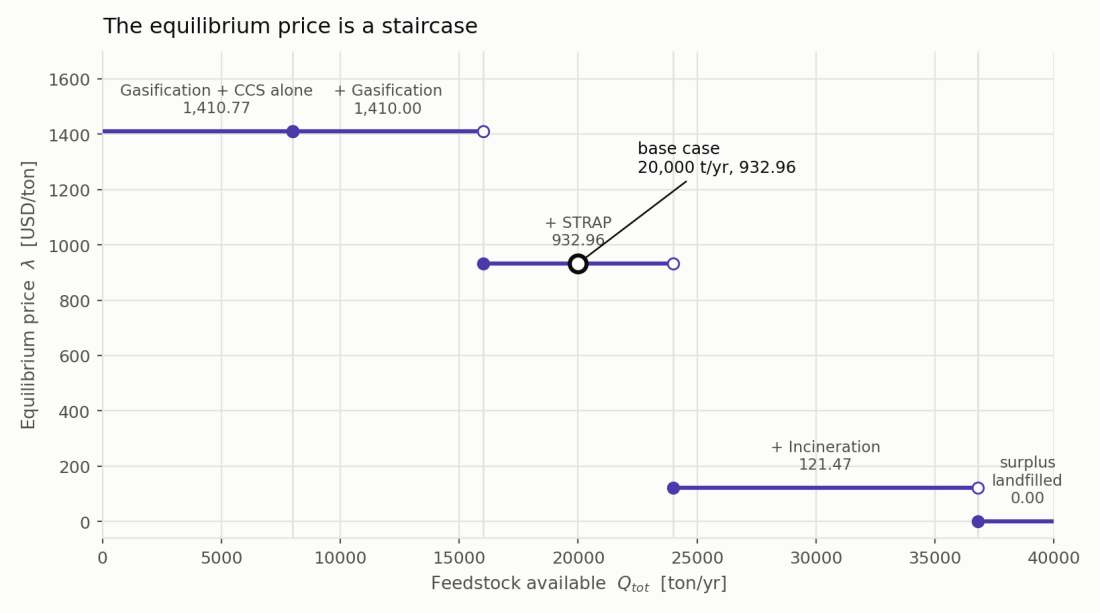


Figure 3: Equilibrium price against the amount of waste: a staircase, flat within a step with a jump each time the price-setter changes. Filled dots are the reported (lowest) price at each threshold.

Table 7: The five steps of the price, from (7).

Waste (t/yr)	Who is buying	λ (USD/t)	Price-setter ($= \lambda$)
below 8,000	Gasification + CCS only	1,410.77	Gasification + CCS, m_{CCS}
8,000 to 16,000	+ Gasification	1,410.00	Gasification, m_{Gas}
16,000 to 24,000	+ STRAP **(base case)**	932.96	STRAP, m_{STRAP}
24,000 to 36,800	+ Incineration	121.47	Incineration, m_{Inc}
above 36,800	all full; surplus landfilled	0	nobody

The thresholds are the running totals K_k of the capacities in queue order: 8,000, 16,000 and 24,000, then $24{,}000 + 12{,}800 = 36{,}800$.

Interpretation

The staircase is the demand curve for waste read sideways. While the pile is smaller than the first plant's capacity, that plant alone absorbs everything and the price is its margin. Once the pile is bigger, the leftover goes to the next firm and the price drops to *that* firm's margin. Once the pile exceeds every plant, the surplus has no buyer and the price is zero. The model gives the step, and the step sets the price. Within a step, a ton more or less changes nothing.

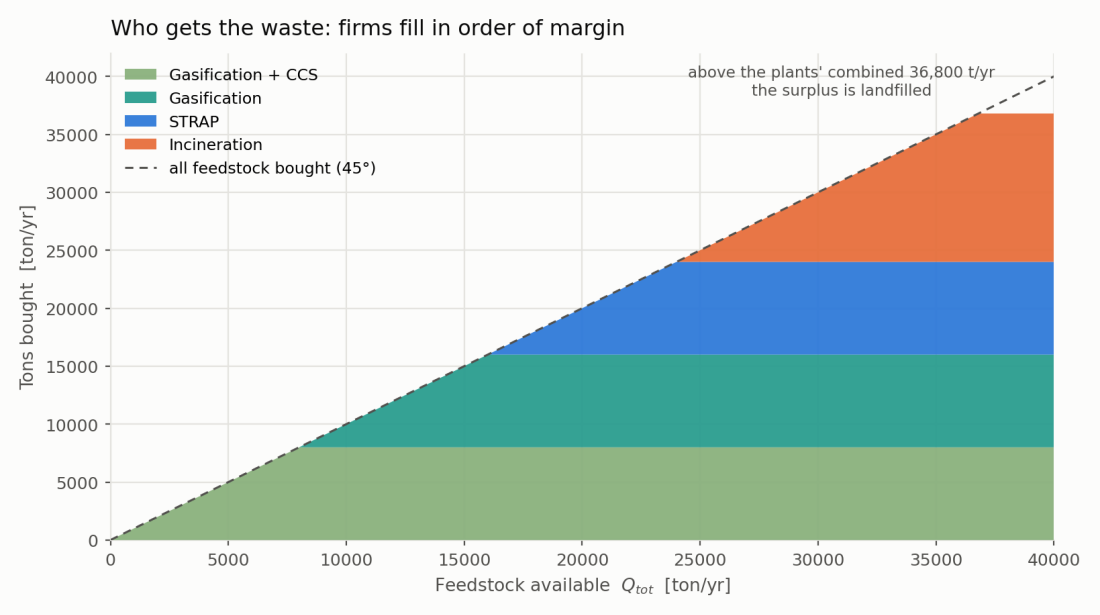


Figure 4: Who gets the waste. Gasification + CCS fills first, then Gasification, then STRAP, then Incineration. Above 36,800 t/yr the gap to the dashed line is waste nobody buys.

The drops are very uneven. The price falls by only 0.76 USD/t at 8,000 (the near-tie between the two gasification routes), by 477 (34%) at 16,000, and by **811 (87%)** at 24,000. The small drop is between the two gasification routes; the large ones are from gasification down to STRAP and from STRAP down to Incineration. The same shows up in the total implicit value of the pile, λQ_{tot}: about 18.7 million USD at 20,000 t/yr, 22.4 million just below 24,000, then 2.9 million just above and 3.6 million at 30,000. A bigger pile can be worth much less in total, because the extra waste goes to a buyer who values it far less.

At the thresholds the price is not unique. At exactly 8,000, 16,000, 24,000 and 36,800 t/yr, any price between the two adjacent steps is consistent with equilibrium. We report the lowest. A drop should be read as "just below versus just above", not as the response to one extra ton.

Interpretation

The base case is not on a knife edge. At 20,000 t/yr we are 4,000 t inside each edge of the STRAP step. The pile could be 20% smaller or 20% larger and the price would still be 932.96, with the same three firms buying.

What this can and cannot tell us. The order of firms and the thresholds depend only on the margins and capacities. The model cannot show a *gradient*: at 17,000 and at 23,000 t/yr it returns the same price. With constant cost per ton it describes scarcity steps, not "a little scarcer". A convex-cost version would give a sloped price and is the natural next step.

3.5 Sensitivity

Why the price depends on STRAP. With these capacities, the best two firms together (16,000 t/yr) cannot cover 20,000 t/yr but the best three (24,000) can. By (7) the third firm in the queue is price-setter:

$$\lambda^* = \text{third-highest margin} = m_{\text{STRAP}} = 932.96.$$

We verified this rule on 1,527 random sets of margins.

Table 8: Effect on λ at 20,000 t/yr of changing one input, all else at the base case. A flip point is where the ranking of firms changes.

Input	Change	Effect on λ
STRAP revenue	+1 USD/t	+1 (one-for-one) while revenue is between 656.90 and 1,945.43 (−55% to +32%). Above 1,945.43, STRAP overtakes Gasification and $\lambda = 1{,}410.00$. Below 656.90, STRAP falls below Incineration and $\lambda = 121.47$.
STRAP operating cost	+1 USD/t	−1 (one-for-one) while cost is between 58.38 and 1,346.92 (−89% to +152%).
Feedstock price p_{raw}	+1 USD/t	−1 (one-for-one); $\lambda = 0$ at 932.96, where STRAP drops out.
Gasification + CCS revenue or cost	any, within limits	None, until its margin falls below STRAP's: revenue below 1,427.19 (−25.1% from 1,905) or cost above 972.04 (+97%). Then $\lambda = m_{\text{CCS}}$, one-for-one. The credit alone cannot trigger this: with no credit at all the margin is still 1,140.77.
Gasification revenue or cost	any, within limits	None, until its margin falls below STRAP's: revenue below 1,157.96 (−29.2%) or cost above 702.04 (+212%).
Incineration revenue or cost	any, within limits	None, unless its margin rises above STRAP's: revenue above 1,071.06, over four times today's 259.57.
Plant capacities	see Figure 6	Jumps between 1,410.00, 932.96 and 121.47.
Capital cost	any	None. Changes profit after capital only.

Interpretation

Three kinds of input. *(i) STRAP's own numbers* move the price one-for-one, because STRAP *is* the price now. A 10% error in STRAP's operating cost (54 USD/t) is a 54 USD/t error in λ, and a 10% error in its revenue (147 USD/t) a 147 USD/t error. *(ii) Capacities* do not shift the price smoothly. They decide which step we are in, so a wrong capacity either changes nothing or changes a lot. *(iii) The gasification and incineration numbers, and the CCS credit,* do not matter at this pile size as long as those firms keep their places in the queue. The flip points say how wrong they would have to be to matter. Gasification + CCS is the most exposed: a 25% drop in its revenue is enough.

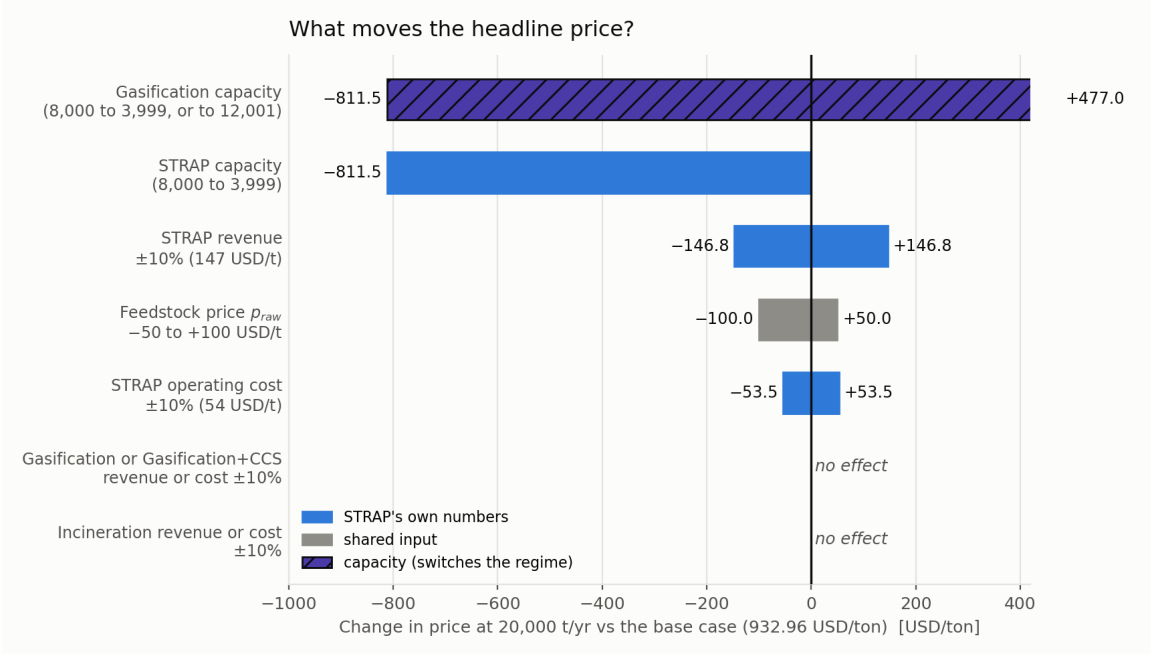


Figure 5: Change in λ at 20,000 t/yr when each input is moved, all else at the base case. Capacities dominate because they switch the step. Among smooth inputs, STRAP's revenue and the feedstock price matter most.

Interpretation

Figure 6 shows how wrong the capacity assumptions can be before the price jumps. From (8,000; 8,000):

- The Gasification capacity can **fall by 4,000** or **rise by 4,000**. Fall further and Incineration becomes the price-setter (λ drops from 932.96 to 121.47). Rise further and Gasification itself becomes the price-setter (λ rises to 1,410.00).
- STRAP's capacity can rise without limit but cannot fall below 4,000; below that, STRAP is full and Incineration sets the price.

STRAP is the price-setter exactly when the two plants ahead of it hold less than 20,000 t/yr together and the three together hold at least 20,000. The capacities are assumed, so this ±4,000 t/yr margin is what to keep in mind when reading the headline.

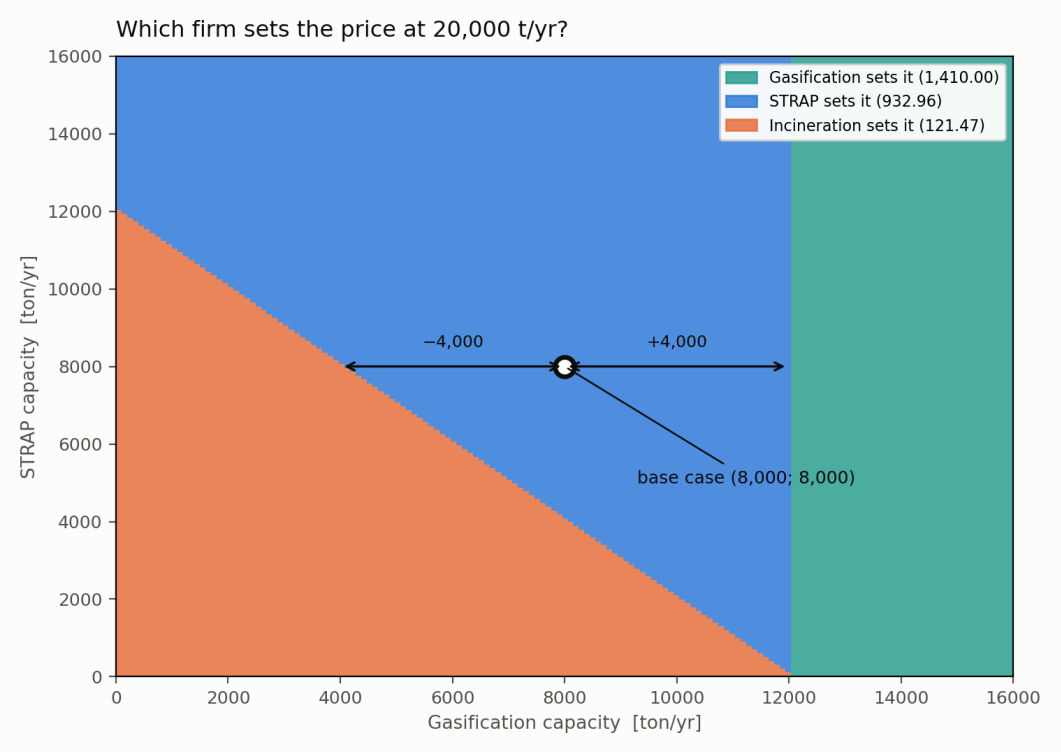


Figure 6: Which firm sets the price at 20,000 t/yr as the Gasification and STRAP capacities vary (Gasification + CCS at 8,000 and Incineration at 12,800 held fixed). The base case is the white dot. Colors come from solving the model at each point.

4 Discussion

4.1 Limitations

- **Linear operating cost.** Cost per ton is constant. This is what makes the price a staircase. A convex-cost version would give a smoothly varying price and a unique split of waste.
- **Fixed prices.** Hydrogen, resin and energy prices are inputs. In reality, if the gasification plants sold much more hydrogen the price would fall, and their margin with it.
- **No cost coupling.** The feedstock price does not depend on how much any firm buys, and landfill is free to the firms. Coupling them would break the equivalence with the planner's problem.
- **Fixed capacities, capital as a check.** Plant sizes are inputs, not the outcome of a capacity-choice game, and capital enters profit but not the equilibrium. Section 3.3 shows STRAP would not break even at these capacities, so a game in which firms choose whether to build would give a different answer. A plant earning capacity rent (here the two gasification plants) also has an incentive to grow.
- **Incineration capital and emissions are missing.** See Q8 and Q9.

4.2 Verification

Every check is in `Code/check.py`; it exits with an error if any fails, and all passed.

1. **Margins** re-typed from Paola's table (with the 270 credit), without the model code, match to 10^{-9}.
2. **Planner LP.** Solving (8) as a linear program gives the same 8,000 / 8,000 / 4,000 / 0 split, with dual values exactly $\lambda = 932.96$, $\mu_{\text{CCS}} = 477.81$ and $\mu_{\text{Gas}} = 477.04$.
3. **Brute force.** For each of the 209 sweep points we scanned candidate prices and took the lowest at which demand fits; it matches the model at all 209. The LP split also matches away from the thresholds.
4. **KKT.** All conditions (2)–(4) hold with a worst violation of exactly zero (the solution is closed-form).
5. **Shape.** The price never rises as the pool grows, takes exactly five values, switches at 8,000 / 16,000 / 24,000 / 36,800 t/yr, and no two firms tie on margin.
6. **Identities.** (9) holds ($26{,}297{,}971 = 18{,}659{,}177 + 7{,}638{,}794$), and the third-highest-margin rule holds on 1,527 random margin sets.
7. **Random problems.** On 3,000 random four-firm problems, the solver's allocation, price and KKT conditions agree with the LP.
8. **Capital.** Profit after capital and break-even tons for the three firms with capital data were recomputed from the raw table, STRAP's break-even pool (22,768 t/yr) matches the solver on either side, incineration's missing capital is reported as missing and not as zero, and adding the capital constants to the planner's objective leaves its allocation unchanged.

These checks show that the code and algebra are right, not that the inputs are. The answer is only as good as STRAP's operating cost, the capacities and the pool size.

4.3 Questions for Paola

Each question lists what we assumed and what the answer would change. The effect is computed at 20,000 t/yr with everything else at the base case.

	What we need from you	What we assumed	Effect on λ
Move the headline price directly			
1	**Size of the waste pool.** How much post-industrial multilayer film is available per year to these four plants?	20,000 t/yr **[ASSUMED]**, chosen so the firms genuinely compete	Steps at 8,000, 16,000, 24,000 and 36,800 t/yr (Table 7). Anywhere in 16,000–24,000 gives 932.96. STRAP covers its capital only above 22,768 t/yr.

	What we need from you	What we assumed	Effect on λ
2	**Plant capacities.** Are the plant sizes right? Is there a defensible single-plant size for STRAP (its 8,000 t/yr plant runs half empty at 20,000 t/yr and loses money), and is 12,800 t/yr the right Incineration capacity?	8,000 each for Gasification, Gasification + CCS and STRAP; 12,800 for Incineration **[ASSUMED]**	Jumps between 1,410.00, 932.96 and 121.47. The base case tolerates ±4,000 t/yr on the Gasification capacity (Figure 6).
3	**STRAP operating cost.** We read 535.43 USD/t as the all-in operating cost including landfilling STRAP's residue, with no capital. Is that right?	535.43 USD/t **[PAOLA]** (our reading of "with landfill")	One-for-one: ±10% gives ∓53.5.
4	**STRAP revenue.** Is 1,468.39 USD/t still right, on the same price basis as the gasification revenue?	1,468.39 USD/t **[PAOLA]**	One-for-one: ±10% gives ±146.8.
5	**Feedstock price.** What does the waste generator charge, if anything? It could be negative (the generator pays to offload it).	0 USD/t **[ASSUMED]**	−1 per dollar; −50 gives 982.96, +100 gives 832.96. Gasification stops covering its capital above 660 USD/t and CCS above 526.
	Change the ranking or the story		
6	**The 45Q credit.** Is 270 USD per ton of *feed* (not per ton of CO_2 captured), how many years does it last, and is it net of transport and storage costs?	270 USD/t added to CCS revenue **[PAOLA]**	None at 20,000 t/yr. CCS and Gasification are tied to within 0.76 USD/t: a credit below 269.24 puts Gasification first. At 8,000–16,000 t/yr it changes the price by at most 0.76.
7	**Hydrogen revenue.** How is 1,635 USD/t built (hydrogen yield × price), and is it the same for the CCS route?	1,635 for both **[PAOLA]**	None unless it falls by 25.1% (CCS) or 29.2% (Gasification).
	Do not change λ, but are needed for a full analysis		
8	**Incineration capital, and the capital basis.** What is Incineration's annualized capital? Also, is the "MM USD" figure per year for the plant sizes quoted, and how does capital change if a plant is a different size (for example STRAP at 4,000 t/yr)?	Incineration: not included. Others: as supplied **[PAOLA]**	None. Changes profit after capital, and whether Incineration is viable when STRAP is not built.
9	**Emissions.** Per-ton process emissions for each technology (with the capture rate for CCS), and the landfill factor for unbought waste.	Not included	None. Needed to report emissions.
10	**Scale of the cost figures.** Gasification and STRAP costs are for an 8,000 t/yr plant and Incineration's for a 100,000 t/yr plant. Is it fair to hold them constant per ton at the capacities used?	Constant per ton **[ASSUMED]**	None while cost per ton is unchanged; a scale effect would change margins.

Interpretation

If there is time for only three questions, ask **3, 2 and 1**: STRAP's operating cost is the price (Q3), and the capacities and pool size decide which step we are in (Q2, Q1) and whether STRAP pays back. Next come Q6 (the credit that holds the two gasification routes level) and Q8 (Incineration capital, needed to finish the viability analysis).

A Reproducing the results

All commands are run from `Paola_Game_Theory/Code/`.

What	File or command
Inputs (including the credit and capital)	`data.py`
Solver, profits and break-even	`solver.py`
Figures 1–6 and `results/sweep_q_tot.csv`	`python plots.py`
All checks in Section 4	`python check.py`

References

Muñoz-Briones, Ikegwu, Martinez, Granger, Van Lehn, Zavala, Huber and Avraamidou. *Decision support for circular economy waste management of mixed plastic streams: A case study on post-industrial multilayer films.*

Global CCS Institute. *U.S. Preserves and Increases 45Q Credit in "One Big Beautiful Bill Act"*. 2025. https://www.globalccsinstitute.com/u-s-preserves-and-increases-45q-credit-in-one-big-beautiful-bill-act/